CT069-3-3 - Database Security Assignment
Assignment Case Study

A company that manages real estate properties, **Green Acres Realty Sdn Bhd**, previously engaged a software house to develop an **Estate Management System (EMS)** for internal use. The system was designed by a team of full-stack developers, handling all aspects including front-end development, back-end development, database architecture, and security deployment.

However, since the entire system was developed by a single team, important aspects such as **access control**, **user privileges**, **auditing**, and **data obfuscation** were not taken into consideration.

As the company has recently expanded, it now plans to establish its own **IT division** and migrate the EMS for future **in-house development**. (Refer to **Appendix I** for details of the initial database structure implemented by the original developers.)

To support this transition, the company intends to form multiple specialized departments within the IT division, including **property management development**, **client portal development**, **analytics**, **database administration**, and others. They have requested that you provide a **solution for the system migration**, with a **primary focus on database security**.

It is important to note that your new clients are **developers from other IT departments**, not end users.

You are required to do the following:

1. Create new database objects such as tables, views, stored procedures, users to control appropriate access
2. Maintain the balance between Confidentiality, Integrity, and Availability
3. Apply all your knowledge from this module.
    - View
    - Stored Procedure
    - Role
    - User
    - Hash
    - Encryption
    - Masking
    - Backups
    - Server auditing
    - Database auditing
    - Trigger (use on both auditing and operation process)
    - Add new table or edit existing table to improve the operation process.
    - Any other techniques you find suitable you may apply for more bonus marks!

Assignment Requirements:

In this assignment you are required to:

- Work in a group of **(4-5)** members. Each member is required to participate in all tasks. All work must be equally distributed among team members.

Deliverables

A. Implementation (40%)

Provide complete steps and/or SQL script to create your database with sufficient data and implement your solutions with inline comments in each section.

- Update the test cases document with relevant sql queries script to test your solution and document the outcome. You are free to add more test cases to the initial test cases document that is provided to you – *save this as DBS_TestCases_<group number>.docx.*

Note: Compress (ZIP) these files and save as Implementation_<your group number>.Zip before uploading to Moodle

- Demo
  A formal presentation is not required for this project. Instead, students are required to produce a video that demonstrates the system's functionalities and overall workflow. Captions may be used without narration; however, if narration is included, English subtitles must also be provided. The video should have a duration of no less than 5 minutes and no more than 15 minutes, and it should be prepared as though it were being presented to an actual client.

B. Documentation (25%)

Project documentation must contain the below sections:

1. Introduction – max 5 pages (10 marks)

- Discuss about this project, what is this assignment about.
- Provide your team's workload matrix (each members contribution in percentage)
- Provide a complete Data Dictionary (updated) for your database/solution with sufficient explanation.

2. Permission Management – max 20 pages (30 marks)

- Provide Authorization Matrix accompanied with a brief explanation on key points.
- Provide and discuss user management solutions that you implemented to address the threats and fulfil the security and permission requirements.

3. Data protection – max 20 pages (30 marks)

- Provide and discuss the Data Classification Matrix accompanied with a brief explanation on key points.
- Provide and discuss the data protection solutions that you implemented to address the threats and fulfil the data protection & classification requirements.

4. Auditing – max 20 pages (30 marks)

- Provide and discuss the Database Security Audit Matrix accompanied with a brief explanation on key points.
- Provide and discuss the audit solutions that you implemented to address the threats and fulfil the auditing and history requirements.

5. Summary

- Provide a summary for your report.

6. References

Note: Save your document as Report_<your group number>.pdf before uploading to Moodle

Additional Note:

i *i) Font size =12, Font type is Times New Roman. Line spacing is 1.5 or 2 spacing.*

ii *ii) Marks will be awarded based on the logical arrangement, accuracy, and detail level.*

Marks Distribution Policy

- All group members will receive the same marks from lecturer. However, if it is verified that a member has made no contribution to the project, that individual will be removed from the group and evaluated separately.
- 90% marks will be given by lecturer for both A. Implementation and B. Documentation. 10% marks will be peer assessment given among group members. The mark given to each member must be unique with the maximum of 5 and minimum of 0.

Appendix 1 – Initial Database Script as Provided by The Developer

```sql
-- 1. Table: Properties
CREATE TABLE Properties (
    PropertyID INT IDENTITY(1,1) PRIMARY KEY,
    PropertyName NVARCHAR(150),
    Address NVARCHAR(255),
    City NVARCHAR(100),
    State NVARCHAR(100),
    Price DECIMAL(18,2),
    Status NVARCHAR(50),  -- e.g. 'Available', 'Sold', 'Rented'
    CreatedDate DATETIME DEFAULT GETDATE()
);

-- 2. Table: Clients
CREATE TABLE Clients (
    ClientID INT IDENTITY(1,1) PRIMARY KEY,
    FullName NVARCHAR(100),
    ContactNumber NVARCHAR(20),
    Email NVARCHAR(100),
    Address NVARCHAR(255),
    RegisteredDate DATETIME DEFAULT GETDATE()
);

-- 3. Table: Agents
CREATE TABLE Agents (
    AgentID INT IDENTITY(1,1) PRIMARY KEY,
    FullName NVARCHAR(100),
    ContactNumber NVARCHAR(20),
    Email NVARCHAR(100),
    CommissionRate DECIMAL(5,2),
    JoinedDate DATETIME DEFAULT GETDATE()
);

-- 4. Table: Transactions
CREATE TABLE Transactions (
    TransactionID INT IDENTITY(1,1) PRIMARY KEY,
    PropertyID INT FOREIGN KEY REFERENCES Properties(PropertyID),
    ClientID INT FOREIGN KEY REFERENCES Clients(ClientID),
    AgentID INT FOREIGN KEY REFERENCES Agents(AgentID),
    TransactionType NVARCHAR(50),  -- e.g. 'Sale', 'Rent'
    TransactionDate DATETIME DEFAULT GETDATE(),
    Amount DECIMAL(18,2)
);

-- 5. Table: MaintenanceRequests
CREATE TABLE MaintenanceRequests (
    RequestID INT IDENTITY(1,1) PRIMARY KEY,
    PropertyID INT FOREIGN KEY REFERENCES Properties(PropertyID),
    RequestDetails NVARCHAR(MAX),
    RequestDate DATETIME DEFAULT GETDATE(),
    Status NVARCHAR(50)  -- e.g. 'Pending', 'In Progress', 'Completed'
);
```